

ECG5000 Anomaly Detection

A Normal-Only Conv1D and LSTM Autoencoder Comparison

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Abstract

This project detects anomalous heartbeats with autoencoders trained only on normal ECG signals. My contribution is a controlled comparison between a Conv1D autoencoder and an LSTM autoencoder, using the same normal-only split and the same robust threshold rule. On the held-out test set, Conv1D-AE gives the best F1 score (0.912), while LSTM-AE gives slightly higher recall (0.997). The results show a clear precision-recall trade-off.

1 Problem and Data

ECG5000 has 5,000 univariate heartbeat sequences, each with 140 time steps and one of five original labels. I treat label 1 as normal and labels 2–5 as anomalous. The given split has 500 training examples and 4,500 test examples. Only normal examples from the training part are used to fit the models. I reserve 20% of these normal training beats for validation with seed 42, and I keep the full original test split for final evaluation.

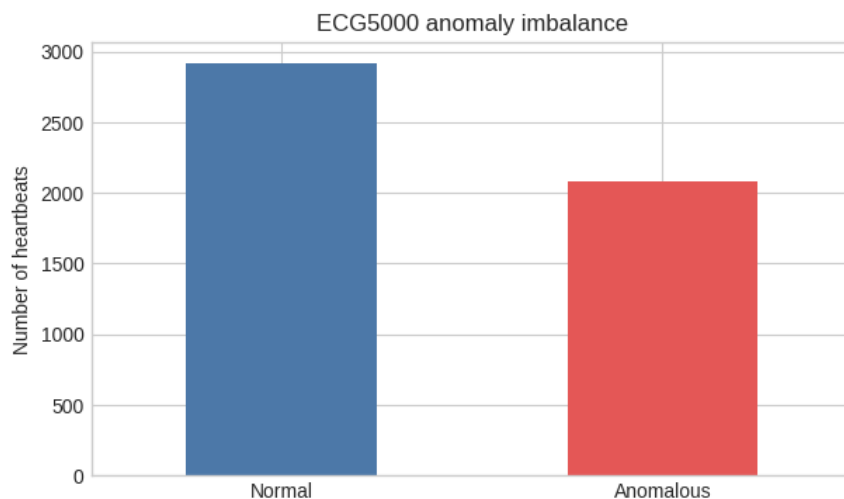


Figure 1: Normal and anomalous beats are imbalanced, so accuracy alone would be misleading.

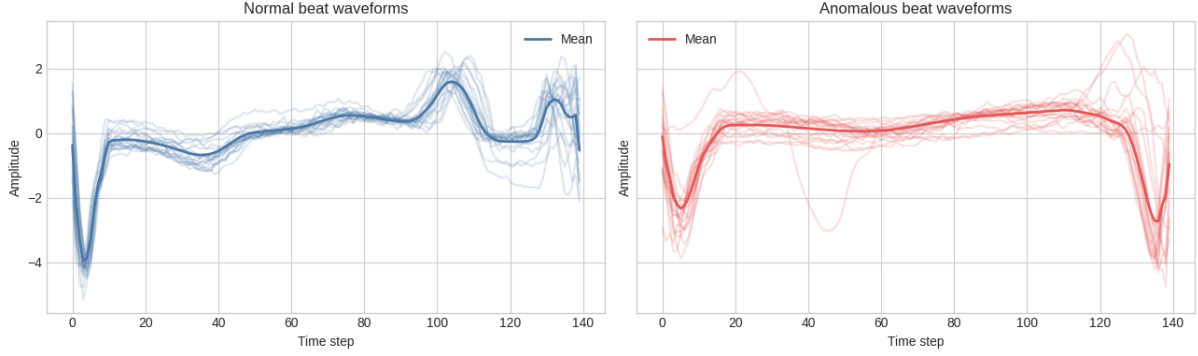


Figure 2: Examples of normal and anomalous waveform distributions.

2 Method and Original Contribution

Both models minimize mean squared reconstruction error on normal beats only. Conv1D-AE uses two strided convolutional encoder stages with transpose-convolution decoding. LSTM-AE uses encoder and decoder LSTMs. Both have a 16-dimensional bottleneck, AdamW, the same batch size, validation split, and seed.

Instead of selecting a fixed percentile, I set each anomaly threshold from the held-out normal validation errors:

$$\tau = \text{median}(e_{val}) + 3 \times 1.4826 \times \text{MAD}(e_{val}).$$

This median absolute deviation (MAD) rule is robust to unusually difficult normal beats. A test beat is anomalous when its reconstruction MSE is above τ . The true anomaly labels are not used for threshold selection.

3 Real Results and Analysis

Table 1 uses the saved output in `outputs/results.csv`. Conv1D-AE has F1 of 0.912, precision of 0.842, and recall of 0.995. LSTM-AE has lower F1 (0.891) and precision (0.805), but it misses only 5 anomalies instead of 10. In contrast, LSTM-AE produces 452 false positives while Conv1D-AE produces 349. For this experiment, Conv1D-AE is the more balanced detector, while LSTM-AE is preferable only if missing an anomaly is more costly than additional alarms.

Table 1: Held-out test results and thresholds from the saved experiment.

Model	Threshold	Precision	Recall	F1	AP	ROC-AUC
Conv1D-AE	1.3548	0.842	0.995	0.912	0.851	0.950
LSTM-AE	1.0754	0.805	0.997	0.891	0.851	0.950

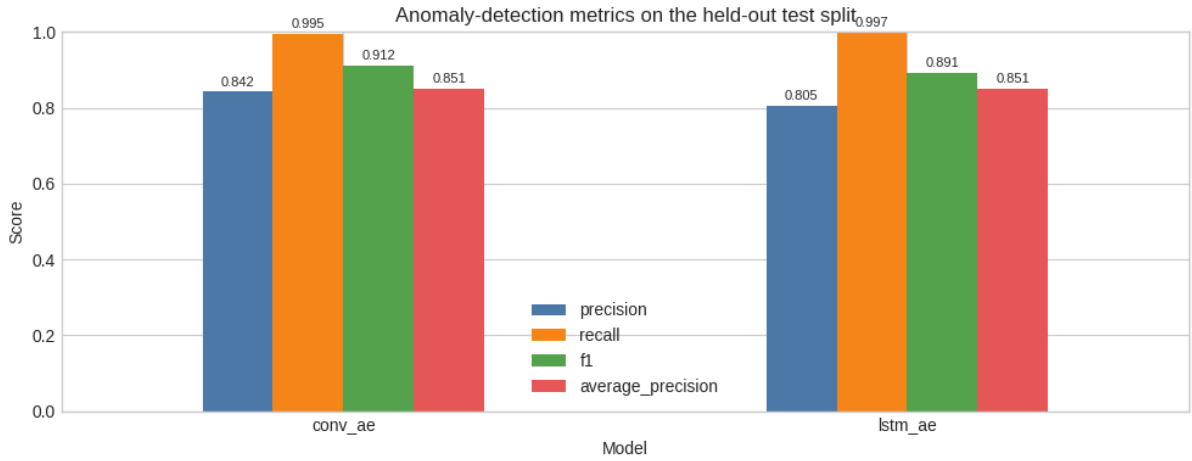


Figure 3: Precision, recall, F1, and average precision for both models.

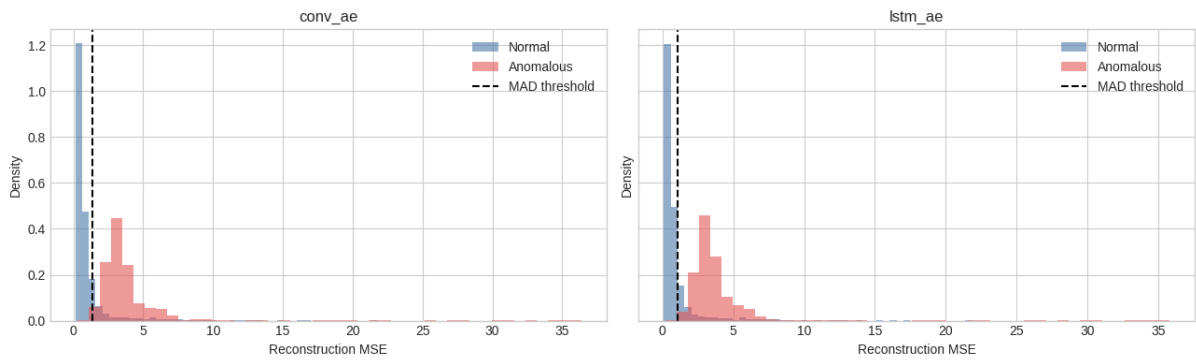


Figure 4: Reconstruction-error distributions and the MAD thresholds.

4 Failure Analysis and Limitations

A normal beat can be falsely flagged when its amplitude or timing pattern is uncommon in the normal-only training subset. A false negative can happen when an anomalous morphology is reconstructed too well. Figure 5 shows saved original/reconstructed examples, which helps to inspect this behavior.

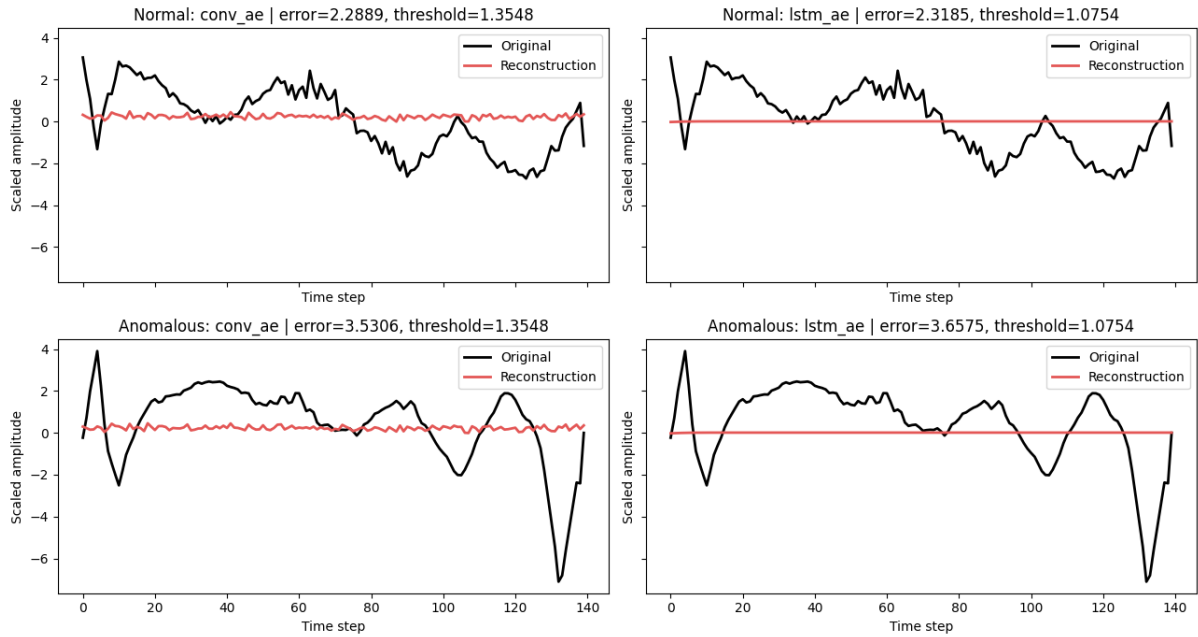


Figure 5: Saved reconstructions for normal and anomalous test beats.

These are one-epoch CPU verification results, so I should not claim that one architecture always wins. Future work should repeat several seeds, report results for each raw anomaly label, and compare with a one-class classical baseline. This would show whether the difference is stable and which morphology types are difficult.

5 Discussion and Observation

The over-capable autoencoder paradox

If an autoencoder has too much capacity, it may learn to reconstruct anomalous beats almost as well as normal beats. Then the reconstruction errors overlap and anomaly detection becomes weak. I try to reduce this risk with a small 16-dimensional bottleneck, shallow models, early stopping, and normal-only training data. A larger model is not automatically better for this task.

Would self-attention help?

Attention could show which time regions are used to form a normal heartbeat representation, for example a peak shape and the recovery region. However, there are only 140 time points and a small normal training set. For this reason, attention may only add parameters and overfit. I would add it only after testing whether the convolutional and LSTM models miss long-range waveform relations.

6 Reproducibility

Run `pythonscripts/train.py` from the project directory. It saves the metrics, error values, model weights, and demo arrays in `outputs/`. Then run the notebooks to recreate the report figures.